

Simplified Machine Provisioning



Agenda

- **Objective:** Review simplified machine provisioning (preview) in Azure Local
- **Scope:** prerequisites, simplified machine provisioning workflow, including installation and registration of Azure Local machines.



Simplified Machine Provisioning

- A new preview feature enabling automated OS installation & registration
- Serves as an alternative to manual installation and registration of Azure Local OS
- Allows machines to be prepared by OEMs, integrators, or customers
- Greatly reduces onsite expertise required

Provision machines with Azure Arc

Secure & simple machine set up. [Learn more](#)



1. Download and install

Download the maintenance environment, image your machine, and collect the voucher.

[View downloads](#)



2. Provision machines

Create Azure resources representing your machines and provide configuration details.

[Provision](#)



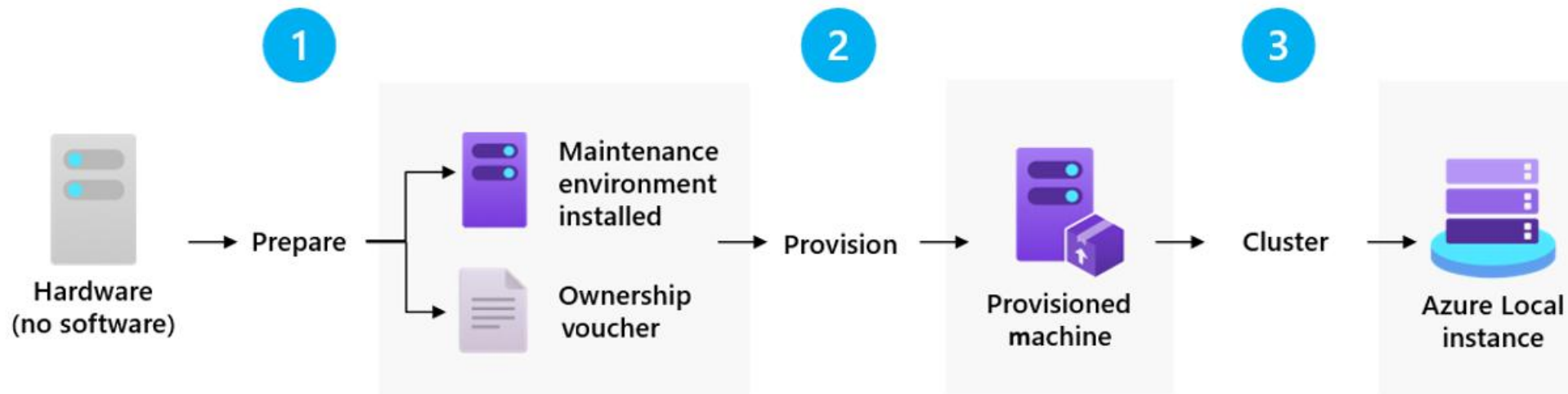
3. Deploy cluster

Deploy an Azure Local instance from the cloud using your provisioned machines.

[Deploy](#)

Provisioning Workflow Overview

1. Prepare the machines
2. Provision from the Azure portal
3. Deploy cluster with provisioned machines



Maintenance environment: lightweight, temporary operating system that boots from a USB drive to securely connect bare-metal hardware to Azure for automated provisioning

Ownership voucher: digital cryptographic file issued by the hardware manufacturer that proves your organization owns the physical server so Azure can safely begin provisioning it

Call-home provisioning: automated step where the booted server contacts a designated Microsoft URL to securely check in, verify its identity using the ownership voucher, and fetch its remote deployment instructions.

Prerequisites

Hardware prerequisites

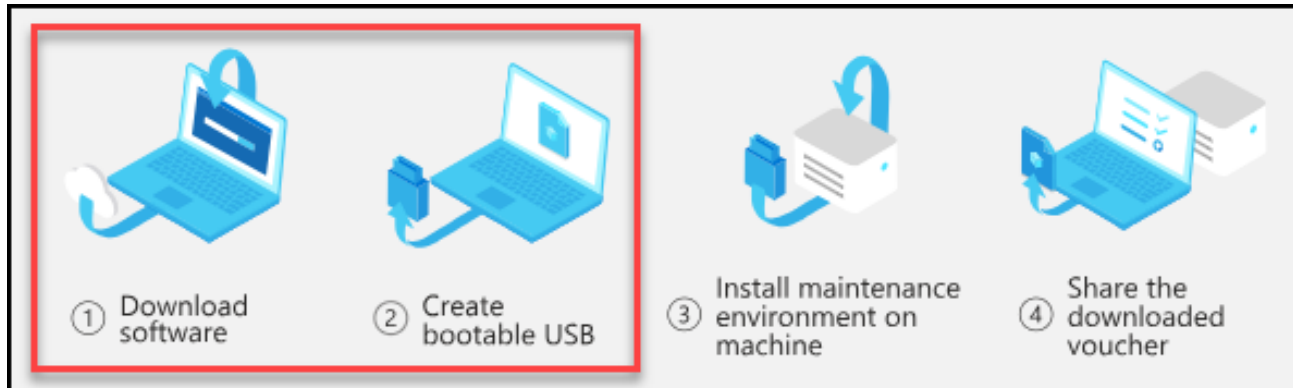
- Microsoft-validated SKUs (Lenovo ThinkAgile MX650, HPE DL360 Gen11, Dell AX-750/650)
- USB port
- Windows 11 PC with access to the internet
- 8GB+ USB drive

Software prerequisites

- Deployment prerequisites including Active Directory preparation
- Downloaded required software (maintenance ISO, USB tool, Configurator app)

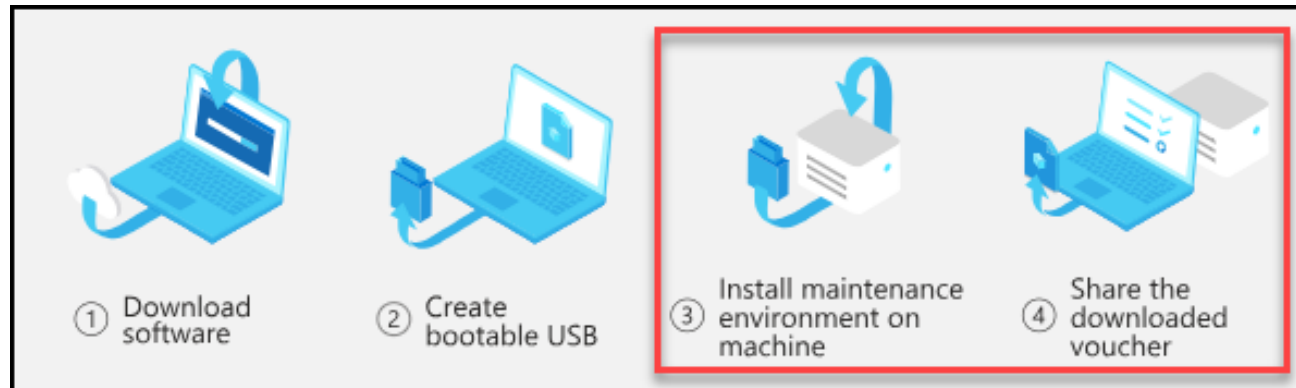
Step 1: Create USB Installation Media

- Download **USB preparation tool** and use it to generate a bootable USB
- Includes:
 - Maintenance environment ISO
 - FDO onboarding components
 - Configurator tool
- Requires Windows Defender exclusion for **usb_prep.exe**



Step 1: Prepare Each Machine

- Boot the server using the prepared USB
- Maintenance environment installs & reboots twice (~30 minutes)
- After completion retrieve the ownership voucher
 - Download via **Configurator App** (connecting to serialnumber.local with default credentials)
 - Copy from **USB drive** (voucher folder)
- The resulting system is in a pre-OS state awaiting Azure initiated-provisioning



Step 2: Provision Machines from the Azure Portal

- Azure Portal → Azure Arc > Azure Local > Get Started > Try provisioning (preview)
- Create a **site**
- Configure site-wide settings:
 - Time zone, time server, proxy, Key Vault for admin credentials, etc.
- Upload **vouchers**, assign software version, create machine resource names
- Click **Review + Create**

What Happens During Provisioning

1. Machine connects to secure **call-home** URL
2. Azure installs **Azure Stack HCI** OS
3. Applies site-defined configuration
4. Connects machine to **Azure Arc**
5. Installs mandatory Arc extensions
6. Machine status transitions to **Ready to cluster**

Monitor Setup Using Configurator App

1. Run **Configurator App for Azure Local V2** on Windows 11
2. Connect using machine serial number or IP
3. Track progress of:
 - Azure Arc configuration
 - OS installation
 - Final cluster-ready state

Verify Connectivity & Completion

1. Azure Portal → **Azure Arc > Operations > Provisioning (preview)**
2. Check **Provisioned Machines** list
3. Review individual machine status
4. Machine should reach final state **Ready to cluster**

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